

Problem Solving Practice:

1)	Given that $(3 + \sqrt{a})(4 + \sqrt{a}) = 17 + b\sqrt{a}$ find the values of a and b .	2)	Given that $(5 - \sqrt{x})^2 = y - 10\sqrt{2}$ find the values of x and y .
3)	Given that $(c - \sqrt{d})^2 = 39 - 12\sqrt{d}$, find the values of c and d .	4)	
5)	If, $(3 + c\sqrt{5})^2 = d + 24\sqrt{5}$, Work out the values of c and d .	6)	Show that $\sqrt{25a^2} + \frac{2ab\sqrt{75}}{\sqrt{3b^2}}$ is always a multiple of 5, given that a and b are integers.
7)	Show that $\sqrt{2}(c\sqrt{8} - d\sqrt{50})$ is always even when c and d are integers.	8)	Given that d is a prime number, rationalise the denominator of $\frac{5\sqrt{d}}{2 + \sqrt{d}}$
9)	Rationalise the denominator of $\frac{2e + \sqrt{f}}{e - \sqrt{f}}$ where e is an integer and f is a prime number.		
Solve, giving your answer in its simplest form:			
10)	$3\sqrt{3} = \sqrt{3x+2}$	11)	$3x - \sqrt{5} = \sqrt{5x+1}$
12)	$x^2 - 2\sqrt{3}x + 3 = 0$	13)	$2x^2 + \sqrt{7}x - 7 = 0$
14)	$3x^2 + 5\sqrt{5}x = 10$		

Challenge Problems:

15) Given that

$$\frac{a + \sqrt{12}}{2 - \sqrt{3}} = b + 10\sqrt{3}$$

find the values of a and b .

16) Given that

$$\frac{a}{(1 + \sqrt{3})^2} + (a\sqrt{3})^3 = 2 + b\sqrt{3}$$

Find the values of a and b .

Answers:

1)	$a=5, b=7$	2)	$x=2, y=27$
3)	$c=6, d=3$	4)	
5)	$c=4$ $d=89$	6)	$\frac{5a + 2ab \times 5\sqrt{3}}{b\sqrt{3}}$ $= 5a + 10a$ $= 15a = 5(3a)$
7)	$\frac{\sqrt{16}c - \sqrt{100}d}{}$ $= 4c - 10d$ $= 2(2c - 5d)$	8)	$\frac{10\sqrt{d} - 5d}{4-d}$
9)	$\frac{2e^2 + f + 3e\sqrt{f}}{e^2 - f}$		
Solve, giving your answer in its simplest form:			
10)	$x = \frac{9-2\sqrt{3}}{3}$	11)	$x = 2 + \sqrt{5}$
12)	$x = \sqrt{3}$	13)	$x = \frac{\sqrt{7}}{2}$ or $x = -\sqrt{7}$
14)	$x = \frac{\sqrt{5}}{3}$ or $x = -2\sqrt{5}$		

Challenge Problems:

15)	$a=6, b=18$
16)	$a=6, b=18$